



**PRT661
DATA SCIENCE PRACTICE**

**ASSESSMENT 01
GROUP PROJECT PLAN:**

**FORECASTING ASSAULT RATES ACROSS NT
REGIONS TO SUPPORT PUBLIC SAFETY
RESOURCE PLANNING**

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1. Project overview and objectives

This project develops an end to end forecasting system for monthly, region level assault rates in the Northern Territory (NT), designed to strengthen operational planning within NT Police, Fire and Emergency Services (PFES). The work involves three core components:

- i. Building a secure and automated data pipeline covering acquisition, storage, and processing of NT crime datasets along with relevant datasets;
- ii. Developing a robust forecasting model capable of producing interpretable, region-specific monthly predictions; and
- iii. Delivering an interactive dashboard that presents historical trends, forecasts, and risk indicators for PFES operational use. The system ultimately supports PFES by improving resource allocation, enabling early identification of emerging hotspots, and enhancing evidence-based decision-making across the Territory.

The system supports PFES through better resource allocation, early hotspot identification, and evidence-based decision-making.

- Primary users: PFES operational planners and duty officers - shift commanders (patrol scheduling), regional operations managers (cross-station coordination), and strategic planning staff (quarterly staffing levels).
- Secondary users: PFES analysts needing historical trend data for budget/policy submissions.

The dashboard targets non-technical operational staff, prioritising clear visual summaries over raw statistical output.

2. Theme justification

The project's core theme is forecasting monthly assault rates across NT regions, within the crime analysis domain. Forecasting suits this data because it is genuinely time-series: assault counts show month-to-month dependence, seasonal patterns (dry vs wet seasons), and temporal autocorrelation - requiring models that capture temporal structure rather than static methods.

This theme aligns with the five mandatory data-science capabilities:

- Acquisition: pulling data from the NTG open data portal (Northern Territory Government, 2026)
- Storage: secure, structured retention of historical records.
- Processing: cleaning, aggregation, and feature engineering into monthly, region-level series.
- Analytics: the forecasting model itself, capturing trends, seasonality, and uncertainty.
- Visualisation: an operational dashboard showing patterns, predictions, and regional risk signals.

Together, these form a complete end-to-end system supporting PFES's operational planning and decision-making.

This project goes beyond training a single model - it spans the full data science lifecycle:

- Data sourcing & reconciliation: integrating 3 heterogeneous public datasets with differing formats and update cycles

- Storage & cataloguing: designing a secure, structured layer for historical and incoming data
- Data engineering: handling structural breaks in crime-recording and inconsistent regional boundaries
- Domain & ethical judgement: addressing Indigenous data governance and predictive-policing misuse risks to shape model outputs
- End-to-end delivery: a user-facing dashboard, not just a model artifact

The forecasting model is just one component of this pipeline. This system-level scope is what distinguishes the project as data science rather than applied ML.

3. Problem statement

PFES faces sustained resource pressure from high, volatile crime volumes across the NT. Monthly assault counts typically range 750 -1,100 incidents Territory-wide, with notable regional fluctuation (Northern Territory Government, 2026); demographic and alcohol-consumption factors may also contribute. These unpredictable surges strain PFES's finite staffing and patrol capacity (Northern Territory Police Association, 2025).

This volatility directly affects PFES's resource planning. Darwin, Alice Springs, Katherine, and Tennant Creek show distinct, shifting assault patterns, requiring frontline teams to adjust without advance warning. High domestic-violence involvement (often 60 - 70% of total assaults) and significant alcohol involvement further intensify demand on policing, investigations, and interagency coordination (Northern Territory Government, 2026). Without a mechanism to anticipate escalation, PFES remains reactive - reallocating resources only after spikes occur.

This project addresses that challenge with a forecasting system producing monthly, region-level assault predictions, backed by a secure data pipeline and dashboard. This gives PFES anticipatory visibility into hotspots and workload, enabling proactive staffing, patrol scheduling, and earlier interventions.

This aligns with PFES's intelligence-led approach, as seen in Operation Ventura (NTPFES, 2026), and extends it by helping PFES anticipate pressure points before they escalate - supporting more efficient resource use and sustained crime-reduction outcomes.

4. Initial architecture diagram

Table 1: Data sources

Datasets	Publisher	Date published	URL	File format	Rows	Columns
Population estimates for NTG Service Regions (1986 to 2025)	Northern Territory Government	07 March 2024	https://data.nt.gov.au/dataset/nt-government-regions-1986-2025	XLSX	17,280	7

NT Crime Statistics 2020-2023	Northern Territory Government	16 January 2024	NT Crime Statistics November 2023 - Dataset - NTG Open Data Portal	CSV	46037	10
NT Crime Statistics Latest	Northern Territory Government	11 June 2026	https://data.nt.gov.au/dataset/current-nt-crime-statistics-april-2026	CSV	9111	10
Wholesale Alcohol Supply by Quarter, 2023	Department of Tourism and Hospitality	13 November 2025	https://data.nt.gov.au/dataset/wholesale-alcohol-supply-by-quarter-2023	XLSX	29	12
Wholesale Alcohol Supply by Quarter, 2024			https://data.nt.gov.au/dataset/wholesale-alcohol-supply-by-quarter-2024		29	12
Wholesale Alcohol Supply by Quarter, 2025			https://data.nt.gov.au/dataset/wholesale-alcohol-supply-by-quarter-2025		29	12

The NT Crime Statistics dataset was selected because it provides monthly time-series crime data by region and offence type, making it suitable for temporal and spatial trend analysis. NTG population data were used to standardise crime rates per 100,000 population, enabling fair comparisons across regions with different population sizes.

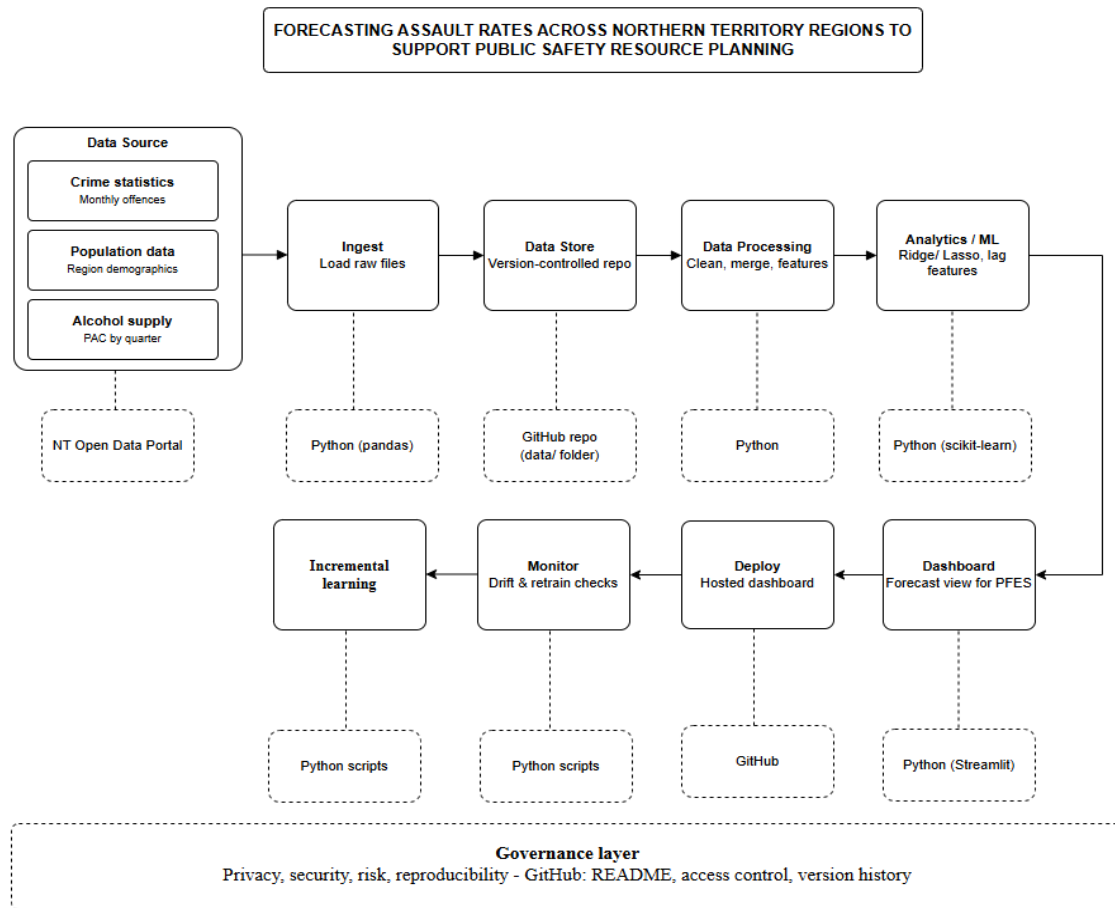


Figure 1: Project Architecture

Table 2: Pipeline stages, tools and purpose

Stage	Tool/ Technology	Purpose
Data Source	NT Open Data Portal	Three public datasets: crime statistics, population demographics, wholesale alcohol supply.
Ingest	Python (pandas)	Loads raw CSV/XLSX files from each source.
Data Store	GitHub repository	Version-controlled storage of raw and processed data, giving change history and access control in place of a cloud data lake.
Data Processing	Python (pandas)	Cleans, merges and engineers features (lagged indicators, cyclical month encoding, per-capita normalisation) across the three sources.
Analytics / ML	Python (scikit-learn)	Trains and validates Ridge and Lasso regression models to forecast monthly, region-level assault rates.
Dashboard	Python (Streamlit)	Presents forecasts and regional trends in an interactive view for PFES.

Deploy	Streamlit Community Cloud (via GitHub)	Automatically redeploys the live dashboard whenever changes are pushed to the GitHub repository.
Monitor	Python scripts	Checks forecast performance for drift and flags when the model should be retrained.
Incremental learning	Python scripts	Continuously adjusting the model parameters by training based on incoming data

Governance considerations spanning the full pipeline (privacy, security and reproducibility) are addressed through GitHub's built-in access controls, commit history and README documentation, and are detailed further in Section 8.

Although the architecture conceptually mirrors AWS's managed data pipeline pattern (S3 → Glue Data Catalog → Glue ETL → SageMaker → QuickSight, with SageMaker Model Monitor for post-deployment monitoring and SageMaker's built-in incremental training capability for scheduled retraining), the group project implements each stage using open-source, non-AWS tools, consistent with the module's guidance to rely on open-source tools for the group deliverable.

Technology roadmap. The current implementation uses open-source tools (pandas, scikit-learn, and Streamlit with GitHub providing version-controlled storage) to deliver a working proof of concept within the semester timeframe. Should PFES adopt the system operationally, the architecture is designed to migrate directly onto AWS-equivalent managed services with minimal redesign: the GitHub-based data store to Amazon S3, pandas-based data processing to AWS Glue ETL, the Analytics/ML stage to Amazon SageMaker for managed training and scheduled deployment, the Streamlit dashboard to Amazon QuickSight, the Monitor stage to Amazon SageMaker Model Monitor, and the Incremental learning stage to SageMaker's built-in incremental (continuous) training capability. Future enhancements under consideration include real-time or near-real-time data ingestion (replacing manual/batch downloads) and integration with additional predictors such as weather or event-calendar data to improve forecast accuracy.

5. Workflow plan

The workflow plan spans the full 12-week semester and is aligned to the four assessment milestones (Table 3). Weeks immediately preceding each submission are used for consolidation and reporting, other weeks are allocated to the team members responsible for the relevant pipeline stage, as shown in Table 4.

Table 3. Twelve-week workflow plan

Week	Focus	Deliverable	Owner
1	Team formation, theme confirmation, environment setup.	GitHub repository with README; Streamlit account set up; draft problem statement	All

2	Data collection and validation, architecture design	Architecture diagram; project management board populated; confirmed data sources	All
3	Proposal writing and initial risk analysis	Assessment 1 submitted: Project proposal and design	All
4	Load raw data into the repository; begin cleaning and harmonising six years of multi-source data; resolve region-mapping and data-regime discontinuities	Raw data in GitHub; cleaned, merged dataset in progress	Will (Ingest) Emma (Data processing)
5	Finalise ingest and storage structure in GitHub; complete data processing scripts; preliminary exploratory analysis	Working ingest/ store/ processing stages; EDA notebook	Will (Ingest, Data store) Emma (Data processing)
6	Consolidate pipeline evidence and baseline model into progress report	Assessment 2 submitted: Progress report and development	All
7	Feature engineering: lagged indicators, cyclical encoding, per-capita normalisation	Feature set finalised for modelling	Hendrick (Analytics/ML)
8	Model training and validation; begin dashboard build	Validated candidate model; dashboard prototype	Hendrick (Analytics/ML) Thomas (Dashboard)
9	Rehearse and deliver technical demonstration	Assessment 3 submitted: Group Technical demonstration and presentation	All
10	Incorporate feedback; finalise dashboard; deploy to Streamlit Community Cloud; implement monitoring and retraining logic	Deployed dashboard; monitoring stage implemented	Thomas (Dashboard, Deploy) Emma (Monitor)
11	End-to-end pipeline testing; draft final report sections	Full pipeline validated; report draft	All
12	Final review and submission	Assessment 4 submitted: Final Professional Report	All

6. Task allocation and team structure

Each team member owns one or two pipeline stages for the duration of the semester (Table 4), ensuring individually attributable contributions are visible in the GitHub commit history across all four assessments.

Table 4. Task allocation and team structure

Member	Primary role	Responsibilities
Will Nguyen	Ingest & Data Store	Loads the three raw datasets and maintains the version-controlled GitHub repository structure (raw and processed data folders)
Emma Tran	Data processing & Monitor	Cleans, merges and engineers features across the three data sources; implements post-deployment drift-checking scripts
Hendrick Dang	Analytics / ML	Develops, tunes and validates the Ridge/Lasso forecasting model
Thomas Tran	Dashboard & Deploy	Builds the Streamlit dashboard and manages its deployment and redeployment via Streamlit Community Cloud.

7. Initial risk analysis

The project has six main risks. These risks will be reviewed during each assessment milestone and updated when needed.

The first risk is changes in the crime data. The transition from PROMIS to SerPro in November 2023 and the ANZSOC classification transition in April 2025 may pose challenges in comparing old and new crime data. To minimise this risk, both transitions will be incorporated into the model.

The second one is that recent crime data may be incomplete because some offences are recorded later. To prevent incorrect results, the latest six months of data will not be included in the model's training and testing.

The third risk is limited data on alcohol supply. This data is only available quarterly and may not cover enough years. The team will collect older data where possible. However, the main forecasting model will still work without alcohol data.

The fourth risk is the same region may be named or bounded differently in different datasets. This can lead to wrong data matching. The team will work with one shared set of regions and validate the data after each merge.

The fifth is a delay in dashboard deployment. To reduce this risk, the team will create and deploy a simple version of the dashboard early on, then improve it throughout the project.

And the final risk is misuse or misinterpretation of the forecasts. If predictions are read as statements about particular communities, the system could cause harm rather than support safety. Small counts in low-population regions could also make incidents identifiable. Forecasts will therefore be shown only at region and month level, with uncertainty ranges, and small values will be suppressed.

8. Ethics, privacy and security considerations

Data privacy

The project only uses public and aggregated data from the Northern Territory Government and the Australian Bureau of Statistics. It does not use personal information or individual records. However, very small crime counts may still make some incidents easier to identify. Small values will therefore be hidden or grouped before being shown on the dashboard.

Indigenous data governance

Aboriginal people are strongly represented in NT crime statistics (Northern Territory Police Force, 2025), so the data must be used carefully and respectfully. The project follows the CARE Principles (Carroll et al., 2020) and the AIATSIS Code of Ethics (AIATSIS, 2020). The model will only make predictions at the regional and monthly level and will not make predictions based on Indigenous status.

Responsible use

The system is designed to help PFES plan staff and resources. It is not designed to predict whether an individual will commit a crime or to target specific addresses or communities.

Forecasts may not always be correct, so the dashboard will show uncertainty and explain the limitations of the data and model. This helps users avoid treating the predictions as exact results.

Security and reproducibility

Only public data and project code will be stored in the GitHub repository. Passwords, API keys and private information will not be uploaded.

The project will also keep its data-processing and analysis steps in version-controlled code. This allows others to understand, replicate, and verify the results.

GitHub repository:

<https://github.com/HendrickDang/PRT661-DATA-SCIENCE-PRACTICE>

Project management board (JIRA):

<https://hendrickdang.atlassian.net/jira/software/projects/SCRUM/boards/1>

9. References

Australian Bureau of Statistics. (2025, June). *National, state and territory population*.
<https://www.abs.gov.au/statistics/people/population/national-state-and-territory-population/latest-release>.

- Australian Bureau of Statistics. (2025). *Prisoners in Australia*. ABS. <https://www.abs.gov.au/statistics/people/crime-and-justice/prisoners-australia/latest-release>.
- Australian Institute of Aboriginal and Torres Strait Islander Studies. (2020). AIATSIS Code of Ethics for Aboriginal and Torres Strait Islander Research. AIATSIS. <https://aiatsis.gov.au/research/ethical-research/code-ethics>
- Carroll, S. R., Garba, I., Figueroa-Rodríguez, O. L., Holbrook, J., Lovett, R., Materechera, S., Parsons, M., Raseroka, K., Rodriguez-Lonebear, D., Rowe, R., Sara, R., Walker, J. D., Anderson, J., & Hudson, M. (2020). The CARE Principles for Indigenous Data Governance. *Data Science Journal*, 19(1), 1–12. <https://doi.org/10.5334/dsj-2020-043>
- Hathaway-Wilson, J. (2025, September 21). *NT police data shows violent crime increasing, property crime decreasing*. Abc.net.au; ABC News. <https://www.abc.net.au/news/2025-09-22/is-nt-crime-decreasing-since-clp-government-elected-police-data/105797920>
- Northern Territory Government. (2026). *NT crime statistics march 2026 - NTG open data portal*. Nt.Gov.Au. <https://data.nt.gov.au/dataset/current-nt-crime-statistics-march-2026>
- Northern Territory Police Association. (2025). Northern Territory Police Association demands better pay and conditions to prevent officer exodus. In *Ntpa.com.au*. <https://www.ntpa.com.au/knowledgebase/article/KA-01198/>
- NTPFES. (2026). *Operation ventura | NT police, fire & emergency services*. NT Police, Fire & Emergency Services. <https://pfes.nt.gov.au/newsroom/2026/operation-ventura>
- Northern Territory Police Force. (2025). *Annual Report 2024-25*. <https://hdl.handle.net/10070/1016397>.